INCH-POUND

MIL-H-6536D
18 April 1994
SUPERSEDING
MIL-H-6536C
22 August 1968

MILITARY SPECIFICATION

HORN AND VALVE, CREW MEMBERS' RELIEF, PRESSURIZED AIRCRAFT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements and tests for two types of horns and valve units.

1.2 Classification. Relief horn and valve units shall be of the following types as specified (see 6.2).

Type I - Straight
Type II - 90-degree, right-angle bend

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Shipping, Fiberboard
PPP-B-676	Boxes, Setup

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-83488	Coating, Aluminum, ION Vapor Deposited

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Technology and Industrial Services Division, SA-ALC/TILDD, 485 Quentin Roosevelt Rd., Kelly AFB, TX 78241-6425 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1680

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STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-831	Test Reports; Preparation of
MIL-STD-970	Standards and Specifications, Order of Preference For The Selection Of

(Unless otherwise indicated, copies of the federal and military specifications, standards and handbooks are available from the Defense Printing Service, Detachment Office, 700 Robbins Ave., Bldg 4D, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 313	Chromium-nickel Stainless and Heat-resisting Steel Spring Wire
ASTM B 36/B36M	Plate Brass, Sheet, Strip, and Rolled Bar

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1137.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article inspection. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.3 Materials.

3.3.1 Brass. The brass used shall conform to ASTM B36/B36M.

3.3.2 Spring steel. The spring used for the release trigger shall be constructed of stainless spring steel, conforming to ASTM A 313.

3.3.3 Aluminum ION. Aluminum ION shall conform to MIL-C-83488, Class 1, or equivalent.

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3.3.4 Plastic. The plastic shall be equal to or interchangeable with lumarith X-T-13507MH, color No. 0-2165-A, Celanese Celluloid Corporation, New York, or Tenite II-201A961MS, Tennessee Eastman Corporation, Kingsport, Tennessee.

3.4 Design and construction. The horn and valve assembly shall consist of a plastic cone-type horn with a spring-tensed valve. The unit shall be so constructed that it can be held and operated by one hand. The valve shall open when the lever is depressed, and shall automatically close when the lever is released. The valve shaft housing shall be large enough to prevent a buildup of negative pressure in the horn chamber when a vacuum is applied across the outlet with the horn chamber plugged.

3.5 Performance.

3.5.1 Valve leakage. When tested as specified (see 4.6.2), the air leakage past the valve seat shall not exceed 0.20 cubic feet per minute (cfm).

3.5.2 Vacuum. With the horn chamber plugged and the valve open, it shall not be possible to induce an absolute pressure of less than 13.7 psia in the horn in 3 minutes' time with the horn outlet connected to a 12.0 psia source.

3.5.3 Valve actuating force. A force not to exceed 1 pound shall be required to open the valve with the lever when a differential pressure of not less than 12 psia is applied to the closed valve (see 4.6.4).

3.5.4 High and low temperature. When tested as specified (see 4.6.5), the horn and valve assemblies shall show no evidence of cracking, chipping, or warping.

3.5.5 Fracture or breakage. The horn and valve assembly shall not break or fracture upon being dropped from a height of not less than 4 feet to a flat metal surface while being maintained at a temperature of 0° ±5° F (see 4.6.6).

3.5.6 Drainage. When tested as specified (see 4.6.7), the time required to drain 50 milliliters of water shall not be more than 3 seconds.

3.6 Dimensions. The type I valve and horn assembly shall conform to the dimensions specified in Figure 1. The type II valve and horn assembly shall conform to the dimensions specified in Figure 2.

3.7 Weight. The weight of the horn and valve assembly shall not exceed 0.20 pound.

3.8 Finish. All non corrosion-resistant steel shall be aluminum ION coated in accordance with MIL-C-83488.

3.9 Identification of product. Horn and valve assemblies shall be marked for identification in accordance with MIL-STD-130 (marking shall not be indent stamped).

3.10 Workmanship. Horn and valve assemblies shall be constructed in accordance with good commercial practice and shall be free from any defect that might affect appearance or functionality.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Sampling instructions. The first article samples shall consist of two horn and valve assemblies representative of the production of horn and valve assemblies, and shall be tested by the contractor.

4.3.2 Tests. First article inspection shall consist of the examination and tests specified (see 4.6).

4.3.3 First article test report. After the contractor completes the first article inspection, a test report shall be prepare in accordance with MIL-STD-831.

4.3.4 First article test sample for the procuring activity. Along with the first article test report, the contractor shall submit one sample to the procuring activity.

4.4 Quality conformance inspection. The quality conformance inspection shall consist of individual tests and sampling tests (see 4.4.1 and 4.4.2).

4.4.1 Individual tests. Each horn and valve assembly shall be subjected to the examination of product specified (see 4.6.1).

4.4.2 Sampling tests. Two horn and valve units shall be selected at random from each lot of 500, or fraction thereof on the order, and subjected to the following tests:

a. Valve leakage (see 4.6.2).
b. Vacuum (see 4.6.3).
c. Valve actuating force (see 4.6.4).
d. High and low temperature (see 4.6.5).
e. Fracture or breakage (see 4.6.6).
f. Drainage (see 4.6.7).

4.4.3 Rejection and retest. When one or more items from a lot fail to meet this specification, acceptance of all items in the lot will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.4.3.1 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot shall not be made until and unless it is determined that the lot meets all the requirements of this specification.

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4.5 Test conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 77° ±18° F and a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowance shall be made for the change in instrument reading.

4.6 Inspection methods.

4.6.1 Examination of product. Each horn and valve assembly shall be examined to determine conformance with the dimensions, workmanship and construction requirements.

4.6.2 Valve leakage. The air leakage past the valve seat shall not exceed 0.20 cfm when the closed valve is subjected to a differential pressure of not less than 12 psia for not less than 3 minutes.

4.6.3 Vacuum. The performance requirement specified (see 3.5.2) is applicable for inspection.

4.6.4 Valve actuating force. The force required to open the valve shall not exceed 1 pound when a differential pressure of not less than 12 psia is applied to the closed valve.

4.6.5 High and low temperature. One sample horn and valve assembly shall be subjected to a temperature of +160° ±5° F for not less than 4 hours. The second sample horn and valve assembly shall be subjected to a temperature of –65° ±5° F, for not less than 4 hours. Any evidence of cracking, chipping, or warping shall be cause for rejection.

4.6.6 Fracture or breakage. The sample horn and valve units shall be dropped from a height of not less than 4 feet to a flat metal surface while the assembly is being maintained at a temperature of 0° ±5° F. Evidence of fracture or breakage shall be cause for rejection.

4.6.7 Drainage. With the horn in a vertical position and the valve fully closed, the horn shall be filled with 50 milliliters of water. Upon opening the valve, the 50 milliliters shall drain from the horn in not more than 3 seconds.

4.7 Packaging inspection. The preservation, packing, and marking shall be inspected to verify conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Horn and valve units shall be preserved and packaged in accordance with MIL-P-116, Method III, one each in a unit container conforming to PPP-B-676.

5.1.2 Level C. Level C shall afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice, provided the latter meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Horn and valve units preserved and packaged as specified (see 5.1.1) shall be packed in overseas type shipping containers conforming to PPP-B-601 or weather-resistant type shipping containers conforming to PPP-B-636. As far as practicable, shipping containers shall be of uniform shape and size, of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each shipping container shall not exceed the weight limitation of the specification. Containers shall be closed and strapped in accordance with the specification and appendix thereto.

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5.2.2 Level B. Horn and valve units preserved and packed as specified (see 5.1.1) shall be packed in domestic type shipping containers conforming to PPP-B-601 or PPP-B-636. Shipping containers shall be of minimum cube and tare consistent with the protection required. As far as practicable, shipping containers shall be of uniform shape and size and contain identical quantities. The gross weight of each shipping container shall not exceed the weight limitation of the specification. Containers shall be closed and strapped in accordance with the specification and appendix thereto.

5.2.3 Level C. Level C shall afford adequate protection at the lowest rate against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice, provided the latter meets the requirements of this level.

5.3 Marking.

5.3.1 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The horn and valve assemblies covered by this specification are intended for use in pressurized aircraft for relief of crew members. They are designed for use with Bracket - Flyer's Relief Tube, Vertical (MS22025), or Bracket - Flyer's Relief Tube, Horizontal (AN8020, No. 2).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Type required (see 1.2).

c. First article requirements (see 4.3.3 and 4.3.4).

d. Levels of preservation and packing (see 5.1 and 5.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title
4.3.3	DI-T-3727A	Test Reports, Preparation of Weapons Delivery Capability Report

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current cleared DID's are cited on the DD Form 1423.

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6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Part or Identifying Number (PIN). The PIN should be constructed as follows:

M6536-1 Type I - Straight

M6536-2 Type II - 90 Degree, right angle bend

6.6 Subject term (key word) listing.

Aluminum ION
Plastic
chamber
leakage
pressure
Temperature
Fracture
breakage
Drainage

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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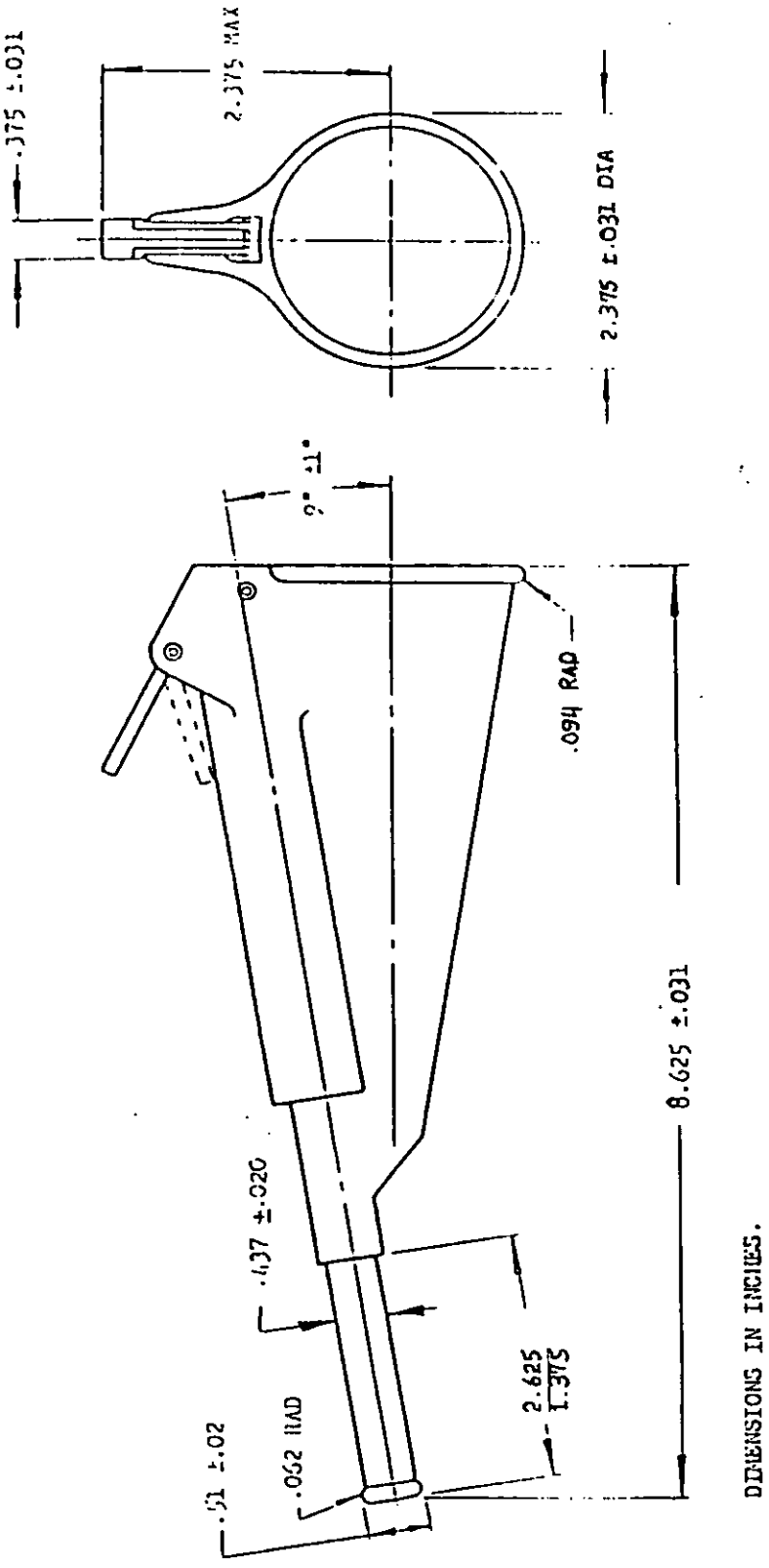


FIGURE 1. Type I horn and valve assembly, fliers' relief tube

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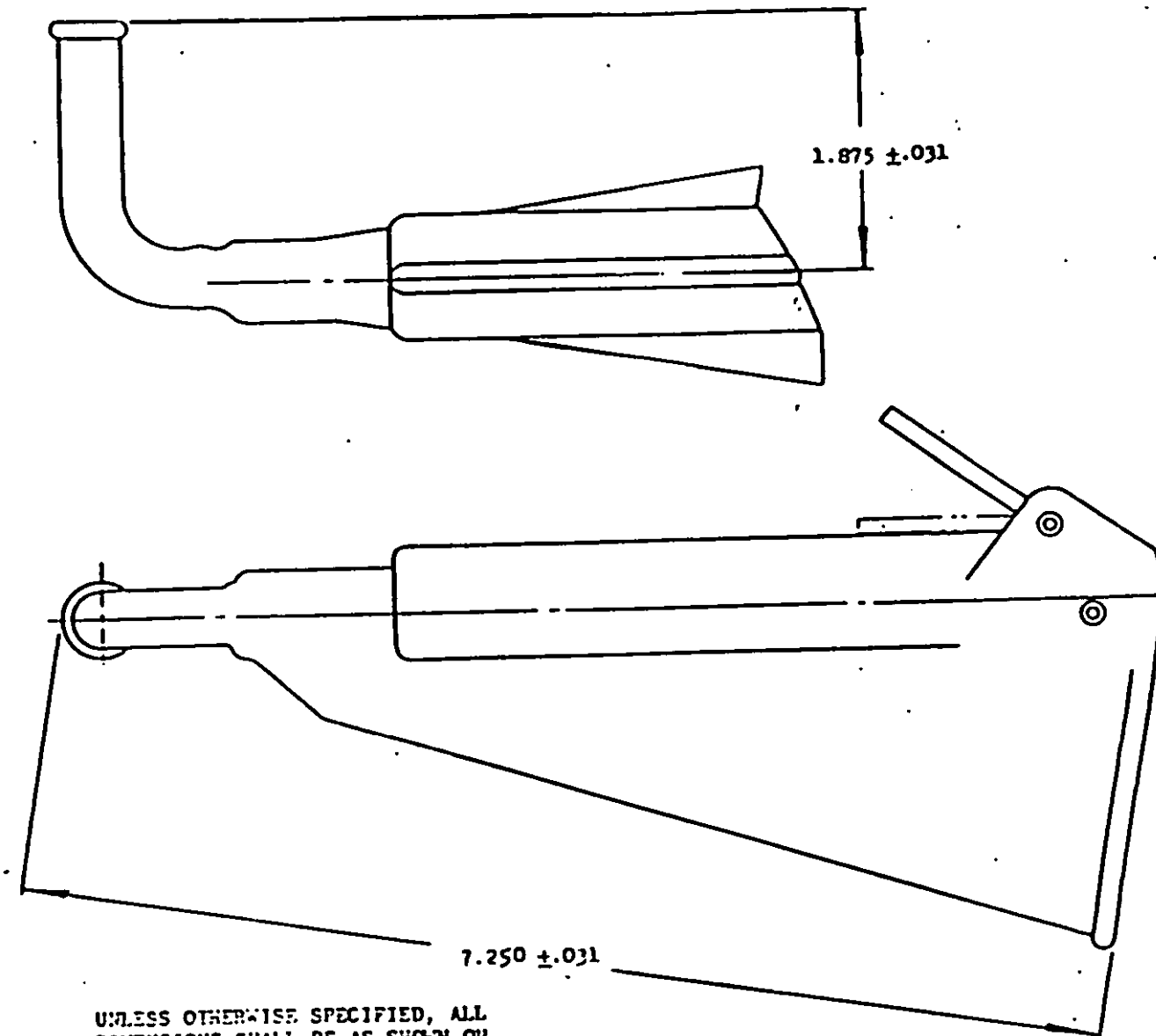


FIGURE 2. Type II horn and valve assembly, fliers' relief tube, 90° right bend

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-H-6536	2. DOCUMENT DATE (YYMMDD) 94 Apr 18

3. DOCUMENT TITLE
HORN AND VALVE, CREW MEMBER'S RELIEF, PRESSURIZED AIRCRAFT

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME MAURIZIO GALBREATH	b. TELEPHONE (Include Area Code) (1) Commercial 616 961-5557	(2) AUTOVON 616 932-5557
c. ADDRESS (Include Zip Code) 74 N. Washington Ave. Battle Creek MI 49017-3094	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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